

Technical Report

Segmentation and Layer Estimation in Graphene Flakes

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1 Introduction

Graphene is a two-dimensional material composed of a single layer of carbon atoms arranged in a hexagonal lattice. Since its discovery, it has attracted great interest due to its electronic, mechanical, and optical properties. One of the main practical challenges in the laboratory is to quickly and reliably identify whether a flake corresponds to a monolayer or to several layers.

A standard technique is **optical microscopy on SiO₂/Si substrates**, where optical contrast allows distinguishing regions with different numbers of layers. The work of Blake et al. (*Making Graphene Visible*, 2007) demonstrated that the visibility of graphene critically depends on the oxide thickness and the wavelength of the light used.

This report describes a computational pipeline developed in Python to:

- Automatically segment graphene regions in optical images.
- Compute relative intensities and contrast between background and flakes.
- Estimate the number of layers present based on experimental contrast.

The complete code is available in the GitHub repository: [JuanGonzalez47/graphene-segmentation](https://github.com/JuanGonzalez47/graphene-segmentation)

2 Theoretical Framework

2.1 Optical Contrast

Contrast is defined as:

$$C = \frac{I_{\text{background}} - I_{\text{flake}}}{I_{\text{background}}}, \quad (1)$$

where $I_{\text{background}}$ is the reflected intensity from the bare substrate and I_{flake} is the intensity from the region covered by graphene.

Blake et al. (2007) reported that, under optimal conditions, typical values are approximately:

- $\sim 6\%$ for 1 layer,
- $\sim 9\%$ for 2 layers,
- $\sim 12\%$ for 3 layers.

This enables distinguishing monolayer from bilayer and few-layer graphene both by eye and through digital analysis.

2.2 Fresnel Laws

Contrast can be explained using the theory of multiple reflection in stratified media. The air/graphene/SiO₂/Si system is modeled applying the **Fresnel equations**, where the reflected intensity depends on:

- refractive indices n_0, n_1, n_2, n_3 ,
- layer thicknesses (d_1 for graphene, d_2 for SiO₂),
- wavelength λ of incident light.

Graphene contributes both by its phase effect and its absorbance. Each additional layer ($d_1 \approx 0.34$ nm) increases absorption and modifies interference, producing discrete steps in contrast.

2.3 Wavelength and Optical Filters

Optical contrast depends on wavelength:

- For ~ 300 nm SiO₂, green light ($\lambda \approx 550$ nm) maximizes contrast.
- For ~ 200 nm thickness, blue light ($\lambda \approx 450$ nm) is more suitable.
- For ~ 100 nm, green or blue can be used.

With white-light microscopy, the camera captures three channels (R, G, B). The green channel is often the most informative, while with a monochromatic filter (e.g., green), only that channel provides useful signal.

2.4 Experimental Image Control

For measured image intensities to be comparable with the model:

- It is recommended to disable automatic white balance and work with RAW or uncompressed images.
- Regions of interest (ROIs) are selected: one on the background (bare Si/SiO₂) and another on the flake.
- Calculations are performed on average intensities in each ROI, avoiding saturation and noise.

Although absolute intensity depends on the camera, the **relative contrast** remains stable and is the quantity compared with theory.

3 Computational Methodology

The developed pipeline follows these steps:

3.1 Channel Selection

The green channel is extracted to maximize contrast, or the channel corresponding to the filter used.

3.2 Illumination Correction

Flat-field correction is applied: the image is divided by a smoothed version (Gaussian blur), removing illumination gradients.

3.3 Multiclass Segmentation

Multi-Otsu thresholding is applied to separate background and several intensity classes (potentially different layers).

3.4 Region Separation

Morphological analysis and watershed are used to separate adjacent or overlapping flakes.

3.5 Contrast and Layer Measurement

For each segmented region, contrast is calculated with respect to the local background. These values are compared with expected contrasts to assign a layer number estimation.

4 Expected Results

The pipeline output includes:

- Image overlay showing segmented regions and layer labels.
- CSV file with properties of each region (area, mean intensity, contrast, estimated layers).
- Intensity histograms and segmentation thresholds.

5 Discussion

- The method is sensitive to illumination conditions and should be applied to unsaturated images.
- Layer estimation requires calibration with samples verified by Raman or AFM.
- Relative contrast is robust to scale changes in the camera, unlike absolute intensity.

6 Conclusions

The proposed pipeline enables:

- Automating the detection and segmentation of graphene flakes in optical microscopy.
- Objectively and reproducibly computing local contrasts.
- Providing a preliminary estimate of the number of layers, optimizing pre-analysis before more costly techniques such as Raman or AFM.

Code Repository: [JuanGonzalez47/graphene-segmentation](https://github.com/JuanGonzalez47/graphene-segmentation)

References

- Blake, P., Hill, E. W., Castro Neto, A. H., Novoselov, K. S., Jiang, D., Yang, R., Booth, T. J., & Geim, A. K. (2007). *Making Graphene Visible*. Applied Physics Letters, 91(6), 063124.
- Discussion notes and analysis developed in this project.